

Same Product, Different Visibility: Representation Robustness in Neural E-Commerce Retrieval

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Abstract

Neural retrieval mediates access to Web content through search, recommendation, retrieval-augmented systems, and emerging agentic interfaces. Semantically equivalent information should ideally receive comparable opportunities to be discovered. Yet structured Web records must often be serialized into ordered token sequences, allowing arbitrary representation choices to affect visibility. We study this problem in e-commerce, where product attributes admit fact-equivalent permutations. Across two benchmarks and three general-purpose retrievers, 4.4–13.3% of highest-relevance pairs cross Top-20 under catalog-wide permutations. A separate target-only intervention holds every competitor fixed and confirms that changing only the target’s serialization can alter its visibility. Reranking reduces instability among surviving candidates but cannot recover items omitted upstream. A simple permutation-invariant set representation eliminates measured order instability, while effectiveness changes are small and statistically inconclusive; this does not prove equivalence. These findings motivate evaluating neural Web retrieval for robustness to semantically irrelevant representation choices alongside relevance.

CCS Concepts

• **Information systems** → **Information retrieval**; *Recommender systems*.

Keywords

Web retrieval, product search, robustness, representation invariance, visibility

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1 Introduction

Web users increasingly reach information through systems that decide what to retrieve before the user ever sees it. Search engines and recommender systems have long mediated this access; retrieval-augmented generation and emerging agentic interfaces add further layers that depend on an initial selection of Web content. This makes retrieval visibility a consequential system property. Ideally, semantically equivalent information should receive comparable opportunities to be discovered: relevance should determine what becomes visible, rather than arbitrary choices in how the same information is represented.

That expectation is difficult to guarantee for structured and semi-structured Web content. Products, jobs, hotels, restaurants, events, and profiles are commonly stored as fields and attribute-value records. Neural retrieval pipelines often flatten those records into text because Transformer encoders consume ordered token

sequences. Many underlying facts, however, have no meaningful linear order. A hotel remains pet-friendly whether that field precedes or follows its parking policy. Once such records are serialized, position and contextual interaction can turn an implementation detail—a template, export order, or serializer—into an unintended retrieval signal. This raises a general Web robustness question: *should semantically equivalent content receive substantially different visibility simply because it is represented differently?*

E-commerce offers a clean testbed for this question. Product attributes such as color, material, dimensions, style, and compatibility are naturally set-like. Their order can be permuted while product identity, semantic content, and human-assessed query relevance remain fixed. We can therefore intervene on representation without asking a model to rewrite, summarize, or add facts. In our controlled comparison, the product is the same, the facts are the same, the query and catalog are the same, and the retrieval model is the same; only the order used to serialize product attributes changes.

Figure 1 makes the consequence concrete. For the query *turquoise chair*, WANDS judges the product Exact-relevant. With all competitors fixed in their original representations, MiniLM ranks the target 5th under one attribute permutation and 1,527th under another. The underlying title and attribute-value multiset are identical. One representation is visible in a top-20 result set, while the other is effectively absent. Why should a serialization choice determine whether the same relevant product can be found?

Query: "turquoise chair"

Same Exact-relevant product and facts
Same model • Every competitor fixed in C0

Order A (C2s4)	Order B (C2s1)
Rank 5	Rank 1,527
VISIBLE at Top-20	HIDDEN at Top-20

Only the target attribute order changes
Should serialization determine visibility?

Figure 1: Target-only intervention: same product, facts, competitors, query, and model. Only the target attribute order changes, moving it from rank 5 to 1,527. All competitors remain in C0; this differs from catalog-wide permutations.

This is more than an embedding-space curiosity. Retrieval systems expose content through operational boundaries: a result page, a recommendation slate, or a candidate pool passed to a later ranker. A score change matters when it moves a relevant item across such

a boundary. If a first-stage retriever excludes the item, a stronger reranker or downstream agent never receives it. Representation sensitivity can thus propagate through a Web retrieval pipeline even when average ranking quality changes little.

Existing product-retrieval research improves lexical, neural, and hybrid matching while usually treating the supplied catalog representation as fixed. Work on generative engine optimization (GEO) instead asks how creators can modify content to gain visibility; e-commerce GEO extends that optimization objective to product descriptions. Our question is prior to strategic optimization: why should fact-equivalent content receive different visibility at all? Recent work on tabular retrieval establishes that alternative serializations can change neural retrieval behavior. We extend this representation-robustness question to item-level product discovery, human relevance, practical cutoffs, candidate-generation loss, and a relevance-aware mitigation.

We study four research questions. **RQ1** asks whether representation alters product visibility across retriever families and product-search benchmarks. **RQ2** asks whether human-relevant products are materially affected, including under direct single-product interventions. **RQ3** asks whether a later cross-encoder can repair the instability or whether first-stage retrieval creates irrecoverable omissions. **RQ4** asks whether an order-invariant representation can improve robustness without a confirmed loss of retrieval effectiveness.

The paper makes four contributions:

- **Problem formulation.** We formalize *representation-induced visibility instability* and connect it to *relevance-visibility alignment*: semantically irrelevant representation changes should not decide which relevant items cross a retrieval boundary.
- **Empirical evidence.** Across two public product-search benchmarks, three general-purpose neural retrievers, and an e-commerce-specific encoder, fact-equivalent attribute orders materially change the Top- K visibility of human-relevant products.
- **Pipeline consequence.** We show that reranking reduces instability among surviving candidates but cannot repair serialization-induced candidate loss.
- **Mitigation.** We evaluate a simple permutation-invariant set representation that eliminates measured order sensitivity while producing a small, statistically inconclusive effectiveness change.

2 Related Work

2.1 Neural Product Retrieval

Product search combines the vocabulary mismatch of Web retrieval with heterogeneous catalog fields and sparse behavioral labels. Lexical methods such as BM25 remain strong components, while neural dual encoders support semantic candidate generation and cross-encoders support more expensive reranking [11, 17, 20, 21]. Hybrid and field-aware systems combine these signals. Choi et al. learn Transformer representations for multi-instance product

fields and explicitly study structured matching together with lexical features [6]; related work extracts query attributes for semantic retrieval and ranking [15]. WANDS and the Amazon Shopping Queries Dataset provide public catalog fields and graded product-search judgments [5, 19]. Dense embedding families and e-commerce-specific encoders broaden the available matching functions [13, 24]. These methods primarily improve matching effectiveness while generally treating the chosen product serialization as fixed.

2.2 Content and Visibility Optimization

A separate line of research studies how content can be changed to gain visibility. GEO evaluates presentational modifications intended to increase a source’s presence in generative answers [2]. E-GEO specializes this objective to product descriptions and generative shopping rankings [3]. SAGEO Arena restores retrieval and reranking to the evaluation boundary and shows why optimization must be assessed across the full search-augmented generation pipeline [12]. Earlier work likewise studies ranking-incentivized, quality-preserving content modification [7].

Our objective differs. GEO asks how content can be modified to obtain more visibility. We ask whether semantically irrelevant representation choices should change visibility at all. The intervention is symmetric: a product can be helped or harmed, and both outcomes count as instability. We neither optimize a target merchant nor infer commercial benefit.

2.3 Representation and Serialization Robustness

Behavioral IR tests reveal model properties that aggregate effectiveness can hide. ABNIRML uses controlled probes for neural rankers [16]; language-model studies show that word order can be simultaneously redundant for some predictions and consequential for others [1, 8, 18]. Robust IR more broadly studies adversarial changes, distribution shift, and performance variance [14].

Structured inputs make serialization a particularly direct concern. TableFormer targets unwanted row- and column-order bias in table reasoning [25]. Most closely related, Bhandari et al. show that semantically equivalent table serializations alter embeddings and retrieval results across retriever families, and study centroid-based representational stability [4]. We therefore do not claim to discover serialization sensitivity in neural retrieval. We extend the question to Web product discovery, where we measure item-level Top- K visibility for human-judged relevant products, distinguish a direct item intervention from a catalog-wide change, quantify candidate-generation loss before reranking, and evaluate robustness jointly with relevance. Ranking-exposure research motivates attention to cutoffs [9, 23], while our evidence remains at the retrieval stage rather than realized user exposure.

3 Problem Formulation

3.1 Representation Robustness for Web Items

Let p be a Web item with underlying semantic information $F(p)$, and let $\mathcal{R}(F(p)) = \{r_1(p), \dots, r_m(p)\}$ be representations that preserve that information for the task. For a query q , human relevance $\text{Rel}(q, p)$ is fixed across this equivalence class. A retrieval model

assigns score

$$z(q, r_i(p)) = \langle E_q(q), E_p(r_i(p)) \rangle. \quad (1)$$

When the differences among r_i are semantically irrelevant, a representation-robust system should ideally make $z(q, r_i(p)) \approx z(q, r_j(p))$, or at least keep the item’s operational retrieval status unchanged. This expectation concerns robustness, not ranking quality: two systems can have similar average NDCG while assigning different visibility to the same relevant item.

We instantiate $F(p)$ as a product’s title, descriptive fields, and attribute-value facts. The equivalence class changes only the order of complete attribute atoms. Query, product identity, facts, catalog, labels, encoder, and ranking rule remain fixed. The induced rank is $\rho(q, p, r_i)$.

3.2 Representation-Induced Visibility Instability

For a cutoff K , the query–item indicator is

$$VI@K(q, p) = \mathbb{I} \left[\min_{r_i \in \mathcal{R}} \rho(q, p, r_i) \leq K < \max_{r_i \in \mathcal{R}} \rho(q, p, r_i) \right]. \quad (2)$$

A value of one is a *Top-K crossing*: at least one equivalent representation makes p visible and another hides it. Pair-micro VI averages over judged pairs. Query-macro VI first averages within each query and then across queries, preventing queries with many judgments from dominating. We report query-bootstrap confidence intervals for the macro estimator. Rank range, $\max_i \rho - \min_i \rho$, describes movement without selecting a cutoff.

Relevance–visibility alignment requires two properties to be assessed together. First, highly relevant items should not cross visibility boundaries because of semantically irrelevant representation changes. Second, a mitigation that improves stability should not obtain that result by degrading relevance. We therefore pair VI with condensed NDCG (cNDCG) [10, 22] and highest-label recall. Condensed metrics remove unjudged products before discounting; complete-catalog VI retains them because they remain competitors for visibility.

For a two-stage system, a relevant item is *irrecoverable at 100* when it appears in the first-stage top 100 under at least one equivalent representation but is absent under another. Reranker stability is evaluated separately on products common to every candidate pool. This avoids crediting a second stage for apparent stability after representation-dependent omissions have already occurred.

3.3 What the Instability Estimator Measures

The equivalence class is finite and operational: it comprises the tested serializations, not every possible rendering of a product record. VI therefore measures whether the observed rank interval straddles a specified cutoff. It is neither the probability that a randomly chosen production serializer will fail nor an estimate over all possible attribute orders. Enlarging the tested family can reveal additional crossings even if the underlying retriever is unchanged. Comparisons consequently keep the same primary family across systems and interventions.

Two features of VI are useful for interpretation. First, the measure is symmetric. A product that becomes visible under a permutation

and a product that becomes hidden both exhibit representation dependence. Calling a crossing a loss relative to C0 would incorrectly treat the source serialization as a privileged semantic reference. Second, VI is local to a boundary. A large rank range entirely below the retrieval cutoff produces no crossing there, whereas a small movement across the cutoff does. Rank range and VI answer complementary questions: how far an item moves, and whether that movement changes its eligibility for retrieval.

Changing the cutoff also changes the question. VI@50 need not exceed VI@20: an item can cross the smaller boundary while remaining visible under every order at the larger one. Conversely, an item invisible under every order at Top-20 may cross Top-50. Thus a cutoff sweep should be read as a profile of operational sensitivity, not as a cumulative failure curve. Reporting several boundaries helps distinguish a result tied to one evaluation convention from a broader pattern of rank instability.

Pair-micro and query-macro aggregation express different populations. The former weights every judged relevant pair equally, so queries with many relevant products contribute more observations. The latter gives each eligible query equal weight after computing its within-query crossing fraction. A difference between the two is not a contradiction or a model-performance gap. It indicates that instability is distributed unevenly across queries with different numbers of judged relevant products. We retain both rather than select the larger estimate as the headline.

Finally, stability alone is insufficient. A retriever that consistently hides every relevant product could have zero VI. Likewise, a constant scoring rule with deterministic ties could be stable but ineffective. Relevance–visibility alignment therefore treats relevance as an essential companion criterion rather than building a single score that could conceal a tradeoff. The empirical question is whether a representation can reduce the nuisance dependence while retaining useful query–product discrimination.

4 Experimental Framework

4.1 Fact-Equivalent Interventions

The primary equivalence family contains seven product representations: the source attribute order, its reversal, and five deterministic per-product permutations. The title and every raw attribute key-value atom are retained. Automatic audits compare atom multiplicity, characters, numeric values, source JSON values, and token multisets; records that fail are excluded before ranking. These are synthetic but fact-preserving interventions: they isolate order rather than attempt to imitate every production template.

The main estimand transforms the complete catalog, reflecting a serializer or indexing-pipeline change. A complementary *target-only permutation intervention* applies the same primary seven-member family to each highest-relevance target independently while keeping every competitor in C0. For variant s , its rank is

$$\rho_s^{\text{target}}(q, p) = 1 + \sum_{u \neq p} \mathbb{I}[z(q, C0(u)) > z(q, s(p))], \quad (3)$$

with ties resolved by original catalog position. The target’s old C0 entry is removed before insertion. We retain per-variant rank changes relative to C0 and compute VI over the seven target-only ranks. This estimates “same product, facts, competitors, query, and

model; only target serialization changes.” The analysis reuses frozen query/product embeddings and variant scores; reconstructed C0 scores and highest-label ranks match stored results exactly in all six model–dataset cells. Earlier single-product M1/M2 template interventions remain complementary tests of broader fact-preserving changes. The first captures system-wide competition; the second isolates the target item’s direct response. They answer different causal questions and need not yield the same rank change.

We also vary chunk length, overlap, and pooling for MiniLM and BGE, while the long-context GTE-ModernBERT provides a further truncation control. These profiles test whether one token budget or pooling rule is sufficient to explain the effect.

4.2 Two-Stage Pipeline Diagnosis

For every query and equivalent serialization, BGE-base retrieves the top 100. The cross-encoder ms-marco-MiniLM-L-6-v2 then scores each query–product candidate. We measure VI before and after reranking for products present in all seven candidate pools, and Irrecoverable@100 over the full highest-relevance population. The distinction is central: reranking can correct ordering among candidates, but it cannot recover content it never receives.

4.3 Proof-of-Concept Invariant Representation

The mitigation follows a structural observation: product attributes are sets, not sentences. We preserve order-sensitive title and description fields in a non-attribute view $n(p)$, encode each raw attribute $a \in A(p)$ independently, and aggregate attributes by their mean. With the same frozen BGE encoder,

$$v(p) = \text{norm} \left((1 - \alpha)E(n(p)) + \alpha \frac{1}{|A(p)|} \sum_{a \in A(p)} E(a) \right). \quad (4)$$

Mean aggregation is permutation invariant, so changing the order of $A(p)$ leaves $v(p)$ unchanged up to floating-point noise. This is a simple proof of concept rather than a claim that set pooling is novel in general. The contribution is its relevance–robustness evaluation for product retrieval.

We select $\alpha \in \{0.25, 0.5, 0.75\}$ on a deterministic 20% WANDS development split, then evaluate the chosen $\alpha = 0.5$ on the remaining 80% and all ESCI queries. Baselines are a factual sentence template (M1), canonical attribute sorting (M2), and late-aggregation ablations. A mitigation is aligned when VI decreases and the paired 95% confidence interval for $\Delta \text{cNDCG}@10$ does not establish a loss; this rule does not prove equivalence.

4.4 Intervention Logic and Controls

The catalog-wide and target-only designs identify different effects. In the catalog-wide design, both a product score and the scores of its competitors may change. This is the appropriate intervention for a shared ingestion template or a catalog reserialization. In the target-only design, every competing score is held fixed. Any rank change must then follow from the target representation and its score relative to that fixed comparison set. This removes competition changes as an explanation for the target’s movement, without assuming that competitors are themselves representation robust.

For target-only evaluation, every relevant pair constitutes a separate counterfactual ranking. We do not simultaneously substitute

all relevant products, which would turn the test back into a catalog-level intervention. For each target, we retain its C0 rank and insert each alternative score into the C0 ranking after removing that target’s former entry. Removing the old entry matters: leaving both copies in the list would create artificial self-competition. Stable tie handling matters for the same reason, particularly in smaller candidate pools. The implementation is checked against explicit stable reranking, including unchanged scores and ties.

Holding competitors in C0 is a controlled reference environment, not a claim that C0 is optimal or universal. The experiment establishes the target effect conditional on that environment. Different competitor representations can change the size or direction of the resulting rank movement. Accordingly, the difference between catalog-wide and target-only VI is not an additive estimate of a separate “competition effect.” Crossing indicators are nonlinear functions of the whole ranked list, and the two interventions can make different pairs cross a boundary.

The representation audit supports this causal interpretation at the input level. Reordering complete attribute atoms leaves their contents and multiplicities intact, and keeps the non-attribute fields fixed. It does not guarantee that a sequence encoder will internally process the variants identically; that is the behavior being tested. Nor does it certify that every arbitrary permutation is equally natural to a human reader. Our intervention is deliberately narrower than a paraphrase or generated rewrite, for which preserving all relevant facts is much harder to verify.

The invariant method changes the architectural treatment of the nuisance variable. Because attributes are encoded independently, an attribute’s vector does not depend on which other attribute appears immediately before it. Commutative pooling then removes dependence on the ordering of those vectors. The non-attribute view remains a sequence, so the method does not discard meaningful word order in titles and descriptions. It is invariant to the specified attribute permutations, not to changes in attribute wording, schema names, values, or the grouping of facts into atoms.

This structure also explains a possible effectiveness cost. Independent encoding removes cross-attribute contextual interaction before aggregation. A useful relation between material, intended use, and dimensions may be easier for a joint encoder to represent than for a simple average. Canonical sorting retains joint encoding but commits to one particular sequence. The empirical comparison therefore concerns two ways of controlling a nuisance variable, each with a different inductive bias; there is no reason to expect either to dominate every dataset.

5 Experimental Setup

5.1 Datasets and Relevance

WANDS contains 42,994 products and 480 evaluated queries after cleaning. We map its Exact, Partial, and Irrelevant judgments to gains 3, 1, and 0. Our Amazon ESCI evaluation uses 500 US test queries selected before retrieval by a seeded SHA-256 order. The shared union catalog contains 10,076 products and 10,134 judgments; E, C, S, and I map to 1, 0.1, 0.01, and 0. The union creates a fixed catalog in which products compete across queries, but most query–product pairs are unjudged. We therefore use condensed

effectiveness metrics and additionally evaluate each query’s official judged candidate pool (median size 16) as a formulation check.

Our primary visibility population contains the highest human-relevance label: Exact in WANDS and E in ESCI. This choice asks whether representation affects items that should clearly be retrievable. Lower relevance strata provide a contrast rather than a definition of harm. The labels express query–product relationship; they are not judgments of objective product quality.

5.2 Retrievers and Ranking Pipeline

The frozen general-purpose retrievers are MiniLM-L6-v2, BGE-base-en-v1.5, and GTE-ModernBERT-base. They span encoder families, mean versus CLS pooling, and native input budgets from 256 to 8,192 tokens. Marqo-Ecommerce-B supplies a domain-specific stress test. Embeddings are normalized and ranked by cosine similarity with deterministic tie handling. The two-stage experiment uses BGE-base for top-100 generation and ms-marco-MiniLM-L-6-v2 as the cross-encoder. Model revisions, tokenizer settings, and hashes are recorded in the artifact.

5.3 Metrics, Inference, and Integrity

We emphasize VI@20 because a twenty-item result set is a plausible visibility boundary, and report VI@10 and VI@50 in the artifact. cNDCG@10 measures judged relevance after condensing unjudged items; highest-label Recall@20 retains complete-catalog positions. Primary uncertainty uses 10,000 query-level bootstrap samples. Method comparisons are paired by query, with prespecified permutation tests and Holm correction where multiple primary contrasts are tested. The confirmatory gate required a query-macro VI@20 confidence-interval lower bound above 1% in multiple model–dataset cells.

The equivalence family, queries, cutoffs, and analysis protocol were frozen before the confirmatory runs. An independent reconciliation recomputed primary metrics from lower-level ranks and matched stored values to a maximum absolute difference of 1.11×10^{-16} . It also found and corrected a join error in an earlier mitigation summary that omitted WANDS queries without Exact items; no primary sensitivity result changed. Query-type labels remain lexical heuristics and are excluded from the main claims.

5.4 Reading the Evaluation Populations

The experiments use three distinct evaluation populations. Primary sensitivity uses all highest-relevance pairs in each complete evaluation catalog. The reranking comparison conditions on membership in every first-stage candidate pool, because ranks after reranking are otherwise undefined for omitted products. Mitigation effectiveness uses the held-out WANDS queries and the ESCI evaluation queries. Values from these populations should not be directly subtracted to infer a treatment effect: even the Original baseline can differ when its query population changes.

Judgment coverage also determines what each metric can support. In the ESCI union, a product judged for one query can compete for another query without having a label there. For visibility analysis it remains a real competitor in the constructed catalog; removing it would change the rank boundary. For condensed effectiveness, unjudged products are removed before computing discounts. This

avoids treating unknown relevance as a known negative, but also means that cNDCG does not fully measure the burden of unjudged products occupying early complete-catalog positions. We therefore report visibility and effectiveness as complementary views.

The official ESCI pool provides a useful diagnostic of this construction. Every candidate there has an official query-specific judgment, but the pool is much smaller than the union catalog. It tests whether instability survives under that alternative ranking population. It does not create a second independent benchmark or an independent effectiveness replication. In particular, when condensing the union removes all additional unjudged competitors, the judged order and resulting NDCG coincide with those of the official pool.

The target-only extension was performed after the confirmatory study. Its role is to strengthen causal isolation using the frozen primary representations, not to redefine the original gate or select new winning configurations. The aggregate estimates cover the same three general retrievers and both datasets rather than only the motivating example. We retain negative and small effects, and label illustrations separately from population summaries. No new model training, relevance annotation, or encoder inference was needed for this extension.

6 Results

6.1 RQ1: Does Representation Alter Product Visibility?

Figure 2 answers the central question across datasets and retriever families. For the three general-purpose encoders, 9.41–13.26% of highest-relevance WANDS pairs and 4.42–8.25% of ESCI pairs cross the top-20 boundary under fact-equivalent attribute orders. Query-macro estimates are larger on WANDS (12.05–19.17%) and remain 3.66–6.91% on ESCI. Every macro confidence interval has a lower bound above the preregistered 1% floor.

The detailed statistics in Supplementary Table S1 show that the WANDS crossings are not merely tie-breaking near rank 20. The median highest-relevance pair moves 81–207 positions across representations. ESCI has a different regime: median movements of two or three ranks for the general encoders still create consistent cutoff crossings because many judged products lie near the boundary. The e-commerce-specific retriever reduces WANDS micro VI@20 to 6.07% but remains at 7.22% on ESCI (Figure 2). Domain training therefore does not make serialization irrelevant.

Two checks narrow possible explanations. First, instability persists in the official ESCI candidate formulation. At top 10, micro VI is 7.60%, 6.65%, and 11.55% for MiniLM, BGE, and GTE; at top 20 it is 3.00%, 2.86%, and 5.89%. Because the median official pool contains 16 products, top-20 crossings are mechanically constrained. Condensed union-catalog NDCG equals official-pool NDCG after unjudged products are removed, so this is a visibility-formulation check rather than an independent effectiveness estimate. Second, changing chunk length, overlap, and pooling does not remove WANDS instability. MiniLM micro VI@20 remains 11.25–13.26% across five profiles, BGE remains 9.41–17.77%, and long-context GTE is also sensitive. A single short-context configuration cannot explain the pattern.

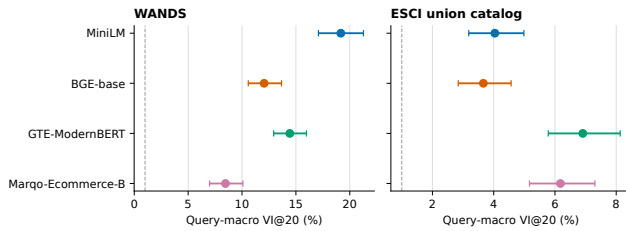
Fact-equivalent serialization changes visibility across retriever families

Figure 2: Equivalent attribute orders change visibility across general and e-commerce-specific retrievers. Error bars are 95% query-bootstrap confidence intervals; the dashed line marks the 1% preregistered floor.

Table 1: WANDS catalog-wide encoding controls. Entries are pair-micro VI@20 percentages. Chunk sizes and overlap are in tokens; pooling specifies the within-chunk rule. These are configuration checks, not separately trained retrievers.

Profile	MiniLM	BGE-base
Native	13.26	9.41
Mean, chunk 126, overlap 32	12.10	14.45
Mean, chunk 254, overlap 0	11.72	17.63
Mean, chunk 254, overlap 64	11.25	17.77
CLS, chunk 254, overlap 0	11.63	17.15

What the encoding controls rule out. Table 1 expands the input-budget and pooling check. Every listed profile retains nonzero highest-relevance crossing, but the profiles do not preserve the same magnitude. In particular, chunking can increase BGE instability even while allowing more of a long record to contribute to its representation. More input coverage should therefore not be equated with order invariance. Window boundaries, within-window context, and aggregation all remain sensitive to how the record is presented.

These controls constrain the explanation without identifying a unique internal cause. The data reject the claim that the observed phenomenon disappears merely by choosing one of these alternative input configurations. They do not decompose the effect into truncation, positional encoding, attention, and pooling contributions. Changing several aspects of an encoding profile is not a clean intervention on any one of those mechanisms. Likewise, the long-context result broadens coverage but does not prove that truncation never matters for any individual record. The supported conclusion is configuration-robust evidence of representation sensitivity, rather than a complete mechanistic account of Transformer order effects.

6.2 RQ2: Are Human-Relevant Products Materially Affected?

The primary results already concern the highest relevance stratum: Exact products in WANDS and E products in ESCI. Figure 1 demonstrates that an Exact item can move by 1,522 ranks while both its facts and all competitor representations remain fixed. At aggregate

scale, WANDS shows more instability among Exact than Irrelevant pairs. The query-macro Exact-minus-Irrelevant difference is 17.01 percentage points for MiniLM, 10.90 for BGE, and 12.62 for GTE; the three 95% intervals exclude zero and Holm-adjusted $p = .0003$. ESCI does not support the same gradient: E-minus-I differences range from -0.41 to $+1.63$ points, all intervals include zero, and adjusted $p \geq .4707$. The defensible cross-dataset conclusion is therefore that highly relevant products are affected, with disproportionate concentration established only on WANDS.

The primary-family target-only test confirms a direct ordering effect with competitors fixed in C0. Micro VI@20 is 12.72%, 8.84%, and 11.39% on WANDS for MiniLM, BGE, and GTE, respectively; corresponding ESCI values are 3.11%, 3.16%, and 6.68%. Query-macro estimates span 11.36–18.96% on WANDS and 2.42–5.88% on ESCI (Supplementary Table S2). These differ from catalog-wide VI because competitors no longer move. Figure 1 uses this stricter intervention: the turquoise chair ranks 5th versus 1,527th against the same C0 competitors, rather than the earlier catalog-wide ranks of 3 and 1,303.

Boundary sensitivity with fixed competitors. Figure 3 reports the same target-only intervention at additional cutoffs. On WANDS, the crossing fraction increases as the evaluated boundary moves deeper, whereas the ESCI rates peak at the smaller reported cutoff and decline thereafter. This difference is compatible with different rank distributions and candidate environments. It should not be read as evidence that a larger result set intrinsically improves or worsens robustness. A boundary can move past the entire rank interval of one product while entering the interval of another.

The target-only median rank range is 210, 83, and 105 on WANDS for MiniLM, BGE, and GTE, respectively; on ESCI it is 0, 1, and 2. The zero ESCI median for MiniLM is compatible with nonzero Top-20 instability: most judged relevant pairs can have stable ranks while a smaller subset crosses the boundary. The aggregate result therefore does not depend on asserting that every product moves substantially. Conversely, the much larger WANDS ranges establish movement beyond adjacent swaps without implying that every such movement reaches a user-visible cutoff.

The motivating example across all tested orders. Table 2 exposes every primary serialization for Figure 1, including those not selected for the opening illustration. The source-order target is outside Top-20, two tested orders put it inside, and another gets close while remaining outside. The direction is not uniformly favorable. This makes the example a test of nuisance dependence rather than a recommended rewriting policy. The table also shows why fixing competitors matters: the same target score can receive a different rank when the surrounding catalog is reserialized.

This case was selected after inspecting the outputs for an intuitive query and a large crossing. It is not an unbiased sample of effect magnitude and cannot substitute for the six aggregate cells. Its scientific role is to make the controlled comparison inspectable: identical facts and fixed competitors can still yield an operationally different result. The population-level inference rests on the full highest-relevance evaluation, not on the visual extremity of this example.

Earlier single-product template interventions provide a complementary check. Replacing only a highest-relevance target with

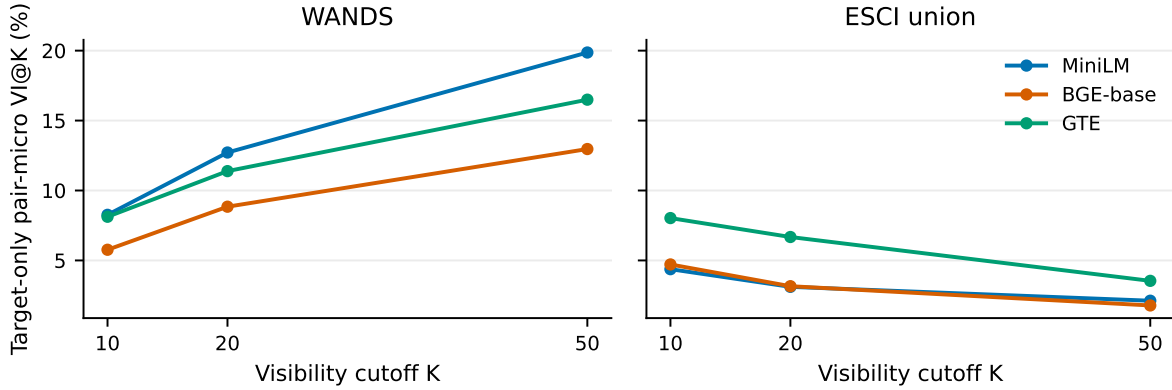


Figure 3: Target-only sensitivity at three visibility boundaries. Each target changes independently; every competitor remains in C0. Points are pair-micro rates, not query-bootstrap estimates. Connecting lines guide the eye and do not interpolate a failure probability.

Table 2: All orders for the Exact-relevant turquoise chair (MiniLM). Target-only competitors stay in C0. Positive Δ means a worse rank than target C0.

Order	Target-only rank	Δ vs C0	Catalog-wide rank
C0	486	0	486
C1	476	-10	295
C2s1	1,527	+1,041	1,303
C2s2	28	-458	21
C2s3	582	+96	484
C2s4	5	-481	3
C2s5	5	-481	4

the fact-preserving M1 or M2 representation yields direct top-20 crossing rates of 8.43–10.27% on WANDS and 7.04–10.80% on ESCI. Transforming the entire catalog yields rates of 6.13–11.54%. Rank directions are mixed: representation changes help some relevant items and harm others. The phenomenon is consequently neither a uniform gain nor merely a side effect of every competitor moving at once.

These results motivate relevance–visibility alignment rather than a new average-effectiveness objective. A system may replace one relevant product with another and retain a similar cNDCG, while the first product’s opportunity to be retrieved changes sharply. VI identifies that item-level instability; relevance metrics prevent a stable but ineffective representation from counting as a solution.

6.3 RQ3: Can a Later Ranking Stage Repair the Effect?

Figure 4 and Table 3 separate within-pool movement from first-stage omission. Among products present in every top-100 candidate pool, the cross-encoder approximately halves query-macro VI@20: from 25.94% to 13.03% on WANDS and from 3.95% to 1.37% on ESCI. Median cross-representation rank range falls from 16 to 8 on

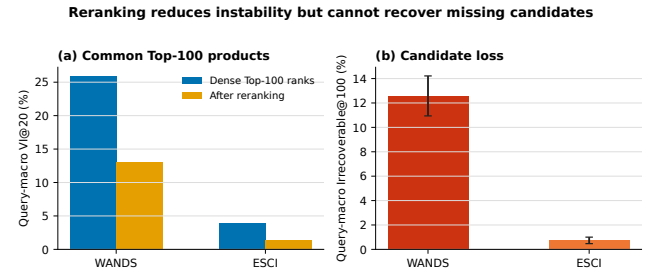


Figure 4: Reranking attenuates instability among common candidates (left), but products omitted by first-stage top-100 retrieval are unavailable to the reranker (right).

Table 3: Two-stage BGE–cross-encoder pipeline. VI is query-macro VI@20 on candidates common to all representation pools; loss is query-macro Irrecoverable@100. All values are percentages.

Data	Dense VI	Rerank VI	Loss (95% CI)
WANDS	25.94	13.03	12.52 [10.94,14.22]
ESCI	3.95	1.37	0.72 [0.47,1.00]

WANDS and from 2 to 1 on ESCI. Reranking can therefore correct some ordering among candidates it sees.

The remaining failure is structural. Under catalog-wide permutations, 15.14% of WANDS highest-label pairs enter the top 100 in some variants but are absent in others, producing a 12.52% query-macro irrecoverable rate. ESCI’s rates are lower (0.99% pair-micro and 0.72% query-macro) but nonzero. Reranking can correct ordering among candidates, but it cannot recover content it never receives. Candidate generation must therefore remain inside the representation-robustness evaluation boundary.

Table 4: Candidate coverage under catalog-wide BGE permutations. “Some only” is ever retrieved minus common to all pools. The conditional loss divides that count by ever retrieved; it is distinct from the all-highest and query-macro rates.

Population	WANDS	ESCI
All highest-relevance pairs	25,470	4,434
Ever in Top-100	12,240	4,039
In every Top-100	8,384	3,995
Some pools only	3,856	44
Loss conditional on ever retrieved	31.50%	1.09%

How many pairs are excluded from the common-pool analysis? Table 4 makes the conditioning explicit. On WANDS, 12,240 highest-relevance pairs appear in at least one top-100 pool, but only 8,384 occur in every pool. The difference, 3,856 pairs, is the representation-dependent candidate-loss population. On ESCI, the corresponding counts are 4,039 and 3,995, a difference of 44. These are query-product pairs rather than unique products: the same catalog item can be relevant to multiple queries.

The conditional rate is useful because it asks about relevant content that the retriever demonstrates it can retrieve under at least one equivalent input. For WANDS this is 31.50%; for ESCI it is 1.09%. These values must not replace the primary all-highest rates without changing the claim. Pairs absent under every serialization are outside this conditional denominator and contribute zero to representation-induced crossing even though their persistent omission can still be a retrieval-effectiveness problem.

There is also a selection effect in evaluating only common candidates. Membership in every pool removes the very products whose availability is unstable at the first-stage boundary. The reduced VI after reranking is therefore evidence about ordering conditional on survival, not a complete end-to-end robustness guarantee. Reporting both candidate coverage and conditional reranker stability prevents this survivorship restriction from being mistaken for recovery of missing content.

For additional context, C0 highest-label Recall@20 rises from 12.58% to 13.61% after reranking on WANDS and from 68.04% to 70.37% on ESCI. These stored C0 point estimates show that improved relevance retrieval and residual representation instability can coexist. They are not paired significance claims, and their cross-dataset difference reflects distinct catalogs and relevance populations. The key pipeline conclusion concerns availability: a later ranker can improve a candidate’s position only when that candidate has reached it.

6.4 RQ4: Can Robustness Improve Without Sacrificing Relevance?

Table 5 compares the set-mean proof of concept with two deterministic baselines. Canonical sorting (M2) mechanically collapses permutations and obtains VI=0, but causes a confirmed held-out WANDS cNDCG@10 decrease of 0.0217. The factual template (M1) leaves residual instability and has a smaller confirmed WANDS decrease of 0.0069. Set mean also obtains VI=0 and changes WANDS

Table 5: Robustness–relevance comparison with BGE. Values are cNDCG@10; parentheses show paired change from Original. WANDS contains the held-out 80%. Macro VI is a percentage.

Data	Method	cNDCG@10 (Δ)	Macro VI
WANDS	Original	.8247	11.93
	M1 factual	.8178 (−.0069)	11.23
	M2 canonical	.8030 (−.0217)	0
	Set mean	.8151 (−.0096)	0
ESCI	Original	.7124	3.66
	M1 factual	.7061 (−.0063)	3.08
	M2 canonical	.7192 (+.0068)	0
	Set mean	.7059 (−.0066)	0

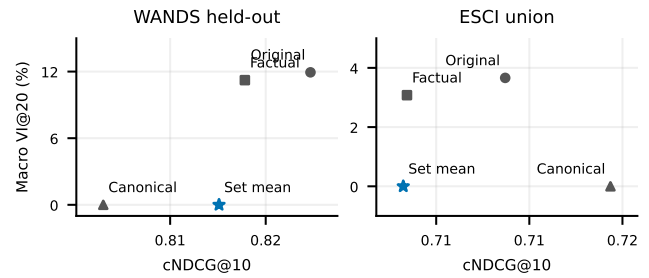


Figure 5: Robustness–relevance frontier. Lower VI and higher cNDCG are preferred. Effectiveness intervals for all variants appear in Figure 6.

cNDCG by -0.0096 , with a paired 95% CI from -0.0195 to 0.0003 . On ESCI, its change is -0.0066 , with a CI from -0.0187 to 0.0055 .

Set mean’s maximum score difference across permutations is 2.38×10^{-7} , below the prespecified 10^{-6} numerical tolerance. The set-based representation thus eliminates measured order-induced instability, while the observed retrieval-effectiveness change is small and statistically inconclusive on both evaluation sets. This does not prove equivalence: the WANDS interval still permits a meaningful loss. The result instead identifies a promising point on a robustness–relevance frontier (Figure 5) and shows why stability and relevance must be evaluated jointly.

Inspecting the effectiveness uncertainty. Figure 6 adds paired confidence intervals for all evaluated mitigation variants. It complements the robustness–relevance scatter plot by exposing the uncertainty that a point-only frontier cannot show. Set mean’s interval crosses zero on both datasets, but much of the WANDS interval lies on the loss side. Calling this “no loss” would conflate failure to establish a difference with evidence that the difference is negligible. A deployment decision would require a prespecified acceptable-loss margin and a study designed to assess that margin.

What the negative variants teach us. All late-aggregation variants satisfy the numerical order-invariance criterion, yet on held-out WANDS their cNDCG@10 values are 0.8058, 0.8093, and 0.8106 for late mean, maximum, and top-3 aggregation. Each paired interval establishes a decrease relative to Original. On ESCI their values are

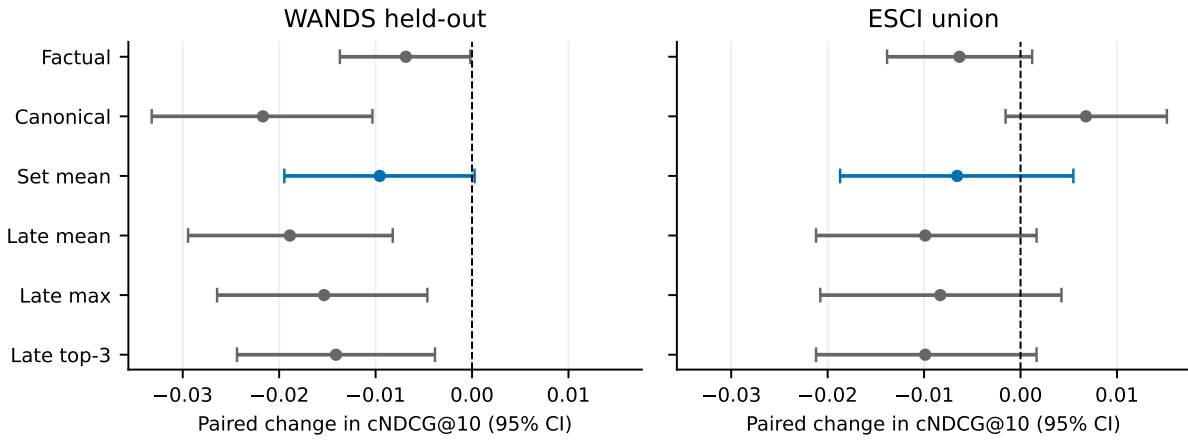


Figure 6: Paired effectiveness changes relative to Original, with stored 95% confidence intervals. The dashed line denotes zero change, not an equivalence margin. Set mean is highlighted; the late-aggregation negative results remain visible.

0.7026, 0.7041, and 0.7026, with intervals spanning zero. Thus invariance does not determine effectiveness: the aggregation rule still controls which aspects of a structured record dominate matching.

Canonical sorting makes the same point from another direction. It obtains zero measured instability but loses effectiveness on WANDS, whereas its ESCI point estimate is higher than Original without a confirmed gain. The available evidence does not establish a universal ranking of the invariant approaches. In particular, a significant Original-versus-canonical contrast and an inconclusive Original-versus-set-mean contrast do not establish that set mean significantly outperforms canonical sorting. Such a claim would require their direct paired comparison.

The useful outcome is therefore a design constraint and a proof of feasibility within the tested family. A representation can remove dependence on attribute order while retaining substantial relevance effectiveness, but the relevance cost must be measured rather than assumed away. Further optimization should preserve this two-objective evaluation, including unsuccessful variants, so that apparent robustness gains cannot be obtained simply by erasing information needed for matching.

7 Discussion

7.1 From Product Retrieval to Web Visibility Robustness

The experiments demonstrate a narrow mechanism with broader relevance. In e-commerce, fact-equivalent serialization is a causal input to whether a human-relevant item crosses a retrieval boundary. Standard relevance aggregates do not fully express this property because one relevant product can replace another while effectiveness remains similar. Representation robustness therefore merits separate evaluation whenever a Web pipeline converts structured content into model input.

The mechanism may arise beyond commerce wherever unordered records are flattened into ordered text: job listings, hotel records, restaurant profiles, event pages, and other catalogs all mix

fields with no canonical semantic sequence. We have not tested those domains and do not claim the same effect or magnitude there. They are plausible settings for future replication because they share the structural precondition, not because the present experiments establish external validity.

For Web indexing, the immediate implication is procedural. Serializers, field templates, and feed transformations should be versioned and regression-tested alongside model weights and index parameters. An evaluation suite can retain equivalence sets for representative records and measure both relevance and Top- K crossing. For multi-stage systems, it should inspect candidate generation explicitly: final-stage evaluation cannot reveal relevant content that never reached the reranker.

Set aggregation illustrates one design response. It preserves sequence where order carries meaning, such as titles and descriptions, while treating schema-defined attributes as a set. The method increases offline indexing cost because attributes are encoded separately, although the resulting product vector remains compatible with standard nearest-neighbor retrieval. Its value here is diagnostic and proof-of-concept. Learned field-aware, set-encoding, or late-interaction models may reach a better relevance-robustness frontier.

7.2 Scope of the Visibility Claim

Retrieval visibility is an upstream opportunity, not realized exposure. Products may belong to merchants, but our data contain no seller groups, impressions, clicks, purchases, revenue, satisfaction, or welfare outcomes. The study therefore does not establish disparate impact or commercial harm. Connecting representation instability to those outcomes requires user logs, merchant meta-data, and an explicit normative model. Keeping this boundary clear strengthens the causal claim we can support: a semantically irrelevant input choice changes which relevant items a retrieval pipeline can surface.

7.3 Limitations

The evidence comes from two public English-language benchmarks and offline retrieval, not a deployed commercial search system. WANDS and the ESCI union have different catalog and judgment structures, which plausibly explains their different rank-displacement regimes. ESCI is incompletely judged; condensed metrics and the official-candidate check address part of this issue but cannot create missing relevance labels.

The permutations are synthetic, although automatic audits verify that they preserve every tested attribute atom, character/token multiset, numeric value, and source value. A blinded human study should additionally assess semantic equivalence and naturalness. Relevance judgments capture query-product relationships rather than objective product quality. We observe no evidence about clicks, sales, purchase decisions, or user satisfaction. VI is also cutoff dependent and does not model examination probability.

Finally, the study covers dense candidate generation and one cross-encoder reranker. It does not establish the same magnitude for sparse, multimodal, generative, or other Web-domain systems. Confidence intervals that include zero for the set-based method do not prove relevance equivalence; the WANDS interval still permits a meaningful loss. A larger prospective non-inferiority test is required before deployment.

7.4 Reproducibility

The artifact records frozen protocols, model revisions, deterministic seeds, equivalence audits, query-level outputs, and source CSVs for every figure. A provenance map links each main-text number to a result field, and an independent audit regenerates primary metrics from stored ranks. No live user or merchant was affected by the experiments.

7.5 An Evaluation Practice for Structured Web Retrieval

The results suggest treating a serializer change as a retrieval-system change, even when it preserves the source record. A useful regression suite would retain query-item relevance judgments together with a small, explicitly defined family of equivalent inputs. Relevance metrics would assess whether the updated pipeline retrieves useful content overall; item-level crossing metrics would assess whether irrelevant formatting decisions determine which particular relevant content survives. Neither assessment substitutes for the other.

The intervention should match the engineering decision. A shared indexing change calls for a catalog-wide test because competitors change together. A record-specific ingestion path calls for a target-only test against a fixed catalog. A multistage service also needs candidate-coverage accounting before any final-ranking comparison. This is a proposed evaluation practice supported by the structure of our findings, not an empirically validated deployment protocol. The appropriate cutoff and tolerance remain application-dependent.

The controlled equivalence definition is equally important. A field list can often be permuted safely, whereas a procedural instruction or a chronological event history may have semantically

necessary order. Applying invariance indiscriminately would destroy useful meaning. Broader Web replication should therefore begin by identifying which components are genuinely unordered for the task and which must remain sequential. Our product setting demonstrates the approach for the audited attribute family; it does not license treating all Web text as a bag of facts.

Finally, representation robustness does not require identical ranking of every relevant item in every context. The expectation is narrower: when the query, facts, and comparison environment are fixed, a nuisance transformation should not arbitrarily determine retrieval eligibility. Personalization, fresh information, changed inventory, and meaningful textual edits are different interventions. Keeping that distinction explicit makes the visibility claim both useful and falsifiable without implying a fairness entitlement or a measured downstream business effect.

8 Conclusion

Neural systems increasingly mediate which Web content becomes visible, yet structured content must often be flattened into ordered text before retrieval. Using e-commerce as a controlled testbed, we show that semantically irrelevant attribute order can change whether human-relevant products cross practical visibility boundaries. The effect reproduces across benchmarks and retriever families, propagates into first-stage candidate loss that reranking cannot repair, and disappears under a proof-of-concept set representation with a small, statistically inconclusive effectiveness change. Web retrieval should therefore evaluate relevance and robustness to semantically irrelevant representation choices together.

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